Acids, Bases and Salts

2 CHAPTER

You have learnt in your previous classes that the sour and bitter tastes of food are due to acids and bases, respectively, present in them. If someone in the family is suffering from a problem of acidity after overeating, which of the following would you suggest as a remedy? lemon juice, vinegar or baking soda solution? Litmus solution is a purple dye, which is extracted from lichen, a plant belonging to the division Thallophyta, and is commonly used as an indicator. When the litmus solution is neither acidic nor basic, its colour is purple. There are many other natural materials like red cabbage leaves, turmeric, coloured petals of some flowers such as Hydrangea, Petunia and Geranium, which indicate the presence of acid or base in a solution.

2.1 UNDERSTANDING THE CHEMICAL PROPERTIES OF ACIDS AND BASES

How do Acids and Bases React with Metals?

Activity 2.1: Take about 5 mL of dilute sulphuric acid in a test tube and add a few pieces of zinc granules to it. What do you observe on the surface of zinc granules? Pass the gas being evolved through the soap solution. Why are bubbles formed in the soap solution? Take a burning candle near a gas filled bubble.

The burning candle burns with a pop sound, indicating the presence of hydrogen gas.

Reaction of zinc with dilute sulphuric acid can be written as follows:

$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$

Notice that the metal in the above reactions displaces hydrogen atoms from the acids as hydrogen gas and forms a compound called a salt. Thus, the reaction of a metal with an acid can be summarised as:

Acid + Metal -> Salt + Hydrogen gas

Can you now write the equations for the reactions you have observed?

2.2 WHAT HAPPENS TO AN ACID OR A BASE IN A WATER SOLUTION?

Do acids produce ions only in aqueous solution? Let us investigate this.

Activity 2.9: Take about 1 g solid NaCl in a clean and dry test tube and set up the apparatus. Add some concentrated sulphuric acid to the test tube. What do you observe? Is there a gas coming out of the delivery tube? Test the gas evolved successively with dry and wet blue litmus paper.

Hydrogen ions in HCl are produced in the presence of water. The separation of H+ ion from HCl molecules cannot occur in the absence of water:
HCl + H2O -> H3O+ + Cl-

Hydrogen ions cannot exist alone, but they exist after combining with water molecules. Thus hydrogen ions must always be shown as H+(aq) or hydronium ion (H3O+).

2.3 HOW STRONG ARE ACID OR BASE SOLUTIONS?

A scale for measuring hydrogen ion concentration in a solution, called pH scale has been developed. The p in pH stands for 'potenz' in German, meaning power.

On the pH scale we can measure pH generally from 0 (very acidic) to 14 (very alkaline). pH should be thought of simply as a number which indicates the acidic or basic nature of a solution. Higher the hydronium ion concentration, lower is the pH value.

The pH of a neutral solution is 7. Values less than 7 on the pH scale represent an acidic solution. As the pH value increases from 7 to 14, it represents an increase in OH- ion concentration in the solution, that is, increase in the strength of alkali.